Advance Data Cleaning

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2026-09-28

Set working directory

```
setwd("C:/Users/ADMIN/OneDrive/Desktop/caleb makini")
```

Import Libraries

```
library(dplyr)
library(tidyverse)
library(rio)
library(readxl)
```

Data importation

```
data = read_excel("dtcln.xlsx")
head(data)
```

```
# A tibble: 6 × 10
    Age Income_KES Household_Size Farm_Size_Acres Monthly_Expenses_KES
  <dbl>      <dbl>          <dbl>           <dbl>                <dbl>
1    58      71751              7            5.74                51513
2    48      67200              4            5.11                31599
3    34      67572              2            4.52                37294
4    NA      58125              1            3.95                30161
5    27      66921              4            3.66                31099
6    40     250000              4            3.21               180000
# i 5 more variables: Years_Experience <dbl>, Education <chr>, Gender <chr>,
#   Extension_Access <chr>, Adoption_Status <chr>
```

Data stracture

```
str(data)
```

```
tibble [100 × 10] (S3: tbl_df/tbl/data.frame)
 $ Age                 : num [1:100] 58 48 34 NA 27 40 58 38 42 30 ...
 $ Income_KES          : num [1:100] 71751 67200 67572 58125 66921 ...
 $ Household_Size      : num [1:100] 7 4 2 1 4 4 1 2 1 6 ...
 $ Farm_Size_Acres     : num [1:100] 5.74 5.11 4.52 3.95 3.66 3.21 0.5 4.41 7.41 4.26 ...
 $ Monthly_Expenses_KES: num [1:100] 51513 31599 37294 30161 31099 ...
 $ Years_Experience    : num [1:100] 13 28 16 29 13 14 3 29 6 18 ...
 $ Education           : chr [1:100] "Secondary" "College" "College" "College" ...
 $ Gender              : chr [1:100] "Female" "Female" "Male" "Male" ...
 $ Extension_Access    : chr [1:100] "No" "Yes" "No" "No" ...
 $ Adoption_Status     : chr [1:100] "Non-Adopter" "Non-Adopter" "Adopter" "Non-Adopter" ...
```

Check for missing values

```
colSums(is.na(data))
```

```
                Age           Income_KES       Household_Size 
                  4                    5                    0 
    Farm_Size_Acres Monthly_Expenses_KES     Years_Experience 
                  3                    3                    0 
          Education               Gender     Extension_Access 
                  4                    3                    3 
    Adoption_Status 
                  2 
```

Separate continuous and categorical variables

```
continuous_vars <- c("Age", "Income_KES", "Household_Size", "Farm_Size_Acres", "Monthly_Expenses_KES",
    "Years_Experience")
categorical_vars <- c("Education", "Gender", "Extension_Access", "Adoption_Status")
head(continuous_vars)
```

```
[1] "Age"                 "Income_KES"          "Household_Size"
[4] "Farm_Size_Acres"     "Monthly_Expenses_KES" "Years_Experience"
```

Check missing values in continuous variables

```
colSums(is.na(data[continuous_vars]))
```

```
                Age           Income_KES       Household_Size
                  4                    5                    0
     Farm_Size_Acres Monthly_Expenses_KES     Years_Experience
                  3                    3                    0
```

Replace missing continuous values with the MEAN

```
for (var in continuous_vars) {
    data[[var]][is.na(data[[var]])] <- mean(data[[var]], na.rm = TRUE)
}
colSums(is.na(data[continuous_vars]))
```

```
                Age           Income_KES       Household_Size
                  0                    0                    0
     Farm_Size_Acres Monthly_Expenses_KES     Years_Experience
                  0                    0                    0
```

Replace missing continuous values with the MEDIAN

```
for (var in continuous_vars) {
    data[[var]][is.na(data[[var]])] <- median(data[[var]], na.rm = TRUE)
}
```

Find the mode for categorical variables

R does not have a simple built-in mode() function for statistical mode, so we create one:

```
get_mode <- function(x) {
    ux <- unique(na.omit(x))
    ux[which.max(tabulate(match(x, ux)))]
}
get_mode(data$Gender)
```

```
[1] "Male"
```

Replace missing categorical values with the MODE

```
for (var in categorical_vars) {
    mode_value <- get_mode(data[[var]])
    data[[var]][is.na(data[[var]])] <- mode_value
}
colSums(is.na(data[categorical_vars]))
```

```
       Education           Gender Extension_Access  Adoption_Status
               0                0                0                0
```

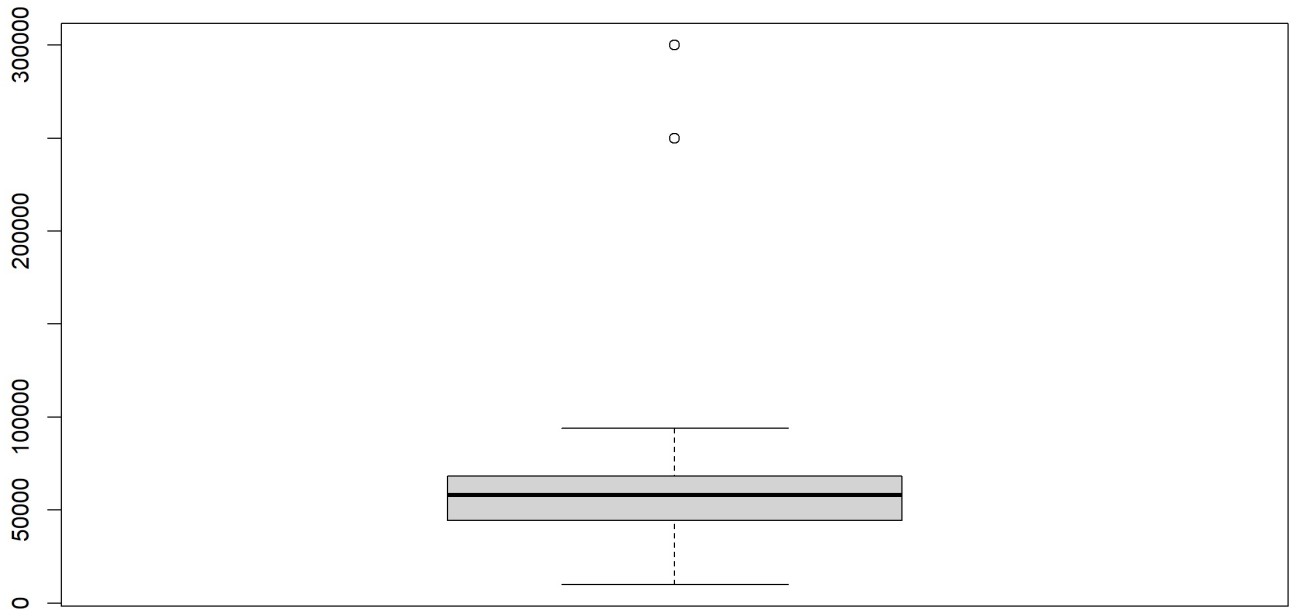
investigate OUTLIERS

```
summary(data[continuous_vars])
```

```
      Age           Income_KES     Household_Size Farm_Size_Acres
 Min.   : 20.00   Min.   : 10022   Min.   : 1     Min.   : 0.500
 1st Qu.: 30.00   1st Qu.: 44522   1st Qu.: 2     1st Qu.: 3.505
 Median : 41.21   Median : 58276   Median : 4     Median : 4.725
 Mean   : 41.42   Mean   : 60219   Mean   : 4     Mean   : 5.854
 3rd Qu.: 48.25   3rd Qu.: 68204   3rd Qu.: 6     3rd Qu.: 6.147
 Max.   :110.00   Max.   :300000   Max.   :25     Max.   :60.000
 Monthly_Expenses_KES Years_Experience
 Min.   :  6357       Min.   : 1.00
 1st Qu.: 24204       1st Qu.: 6.00
 Median : 31527       Median :16.00
 Mean   : 34468       Mean   :15.61
 3rd Qu.: 37284       3rd Qu.:22.25
 Max.   :220000       Max.   :70.00
```

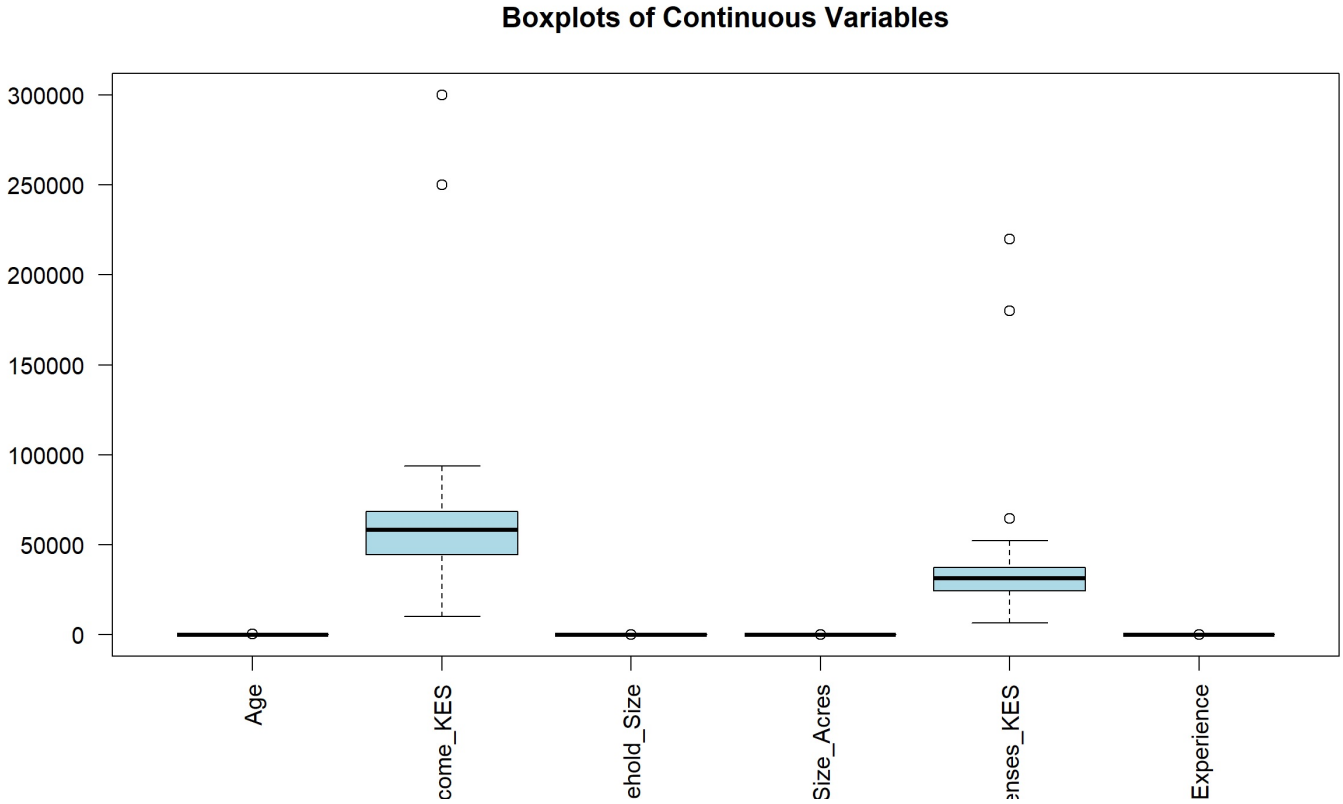
Simple case

```
boxplot(data$Income_KES)
```



Boxplot of all continuous variables

```
boxplot(data[continuous_vars], main = "Boxplots of Continuous Variables", las = 2,
    col = "lightblue")
```



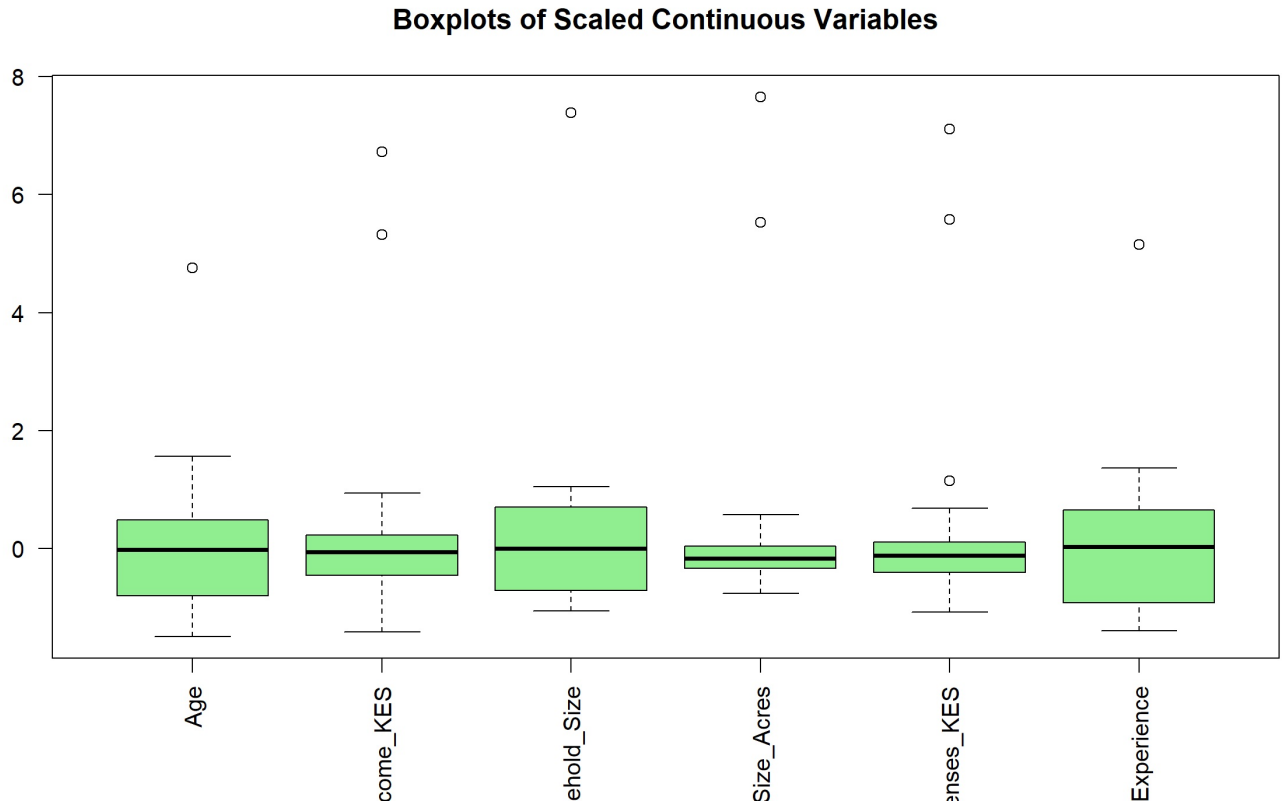
Scale the continuous variables

```
scaled_data <- scale(data[continuous_vars])
scaled_data <- as.data.frame(scaled_data)
head(scaled_data)
```

```
          Age  Income_KES Household_Size Farm_Size_Acres Monthly_Expenses_KES
1  1.14986590  0.32326793      1.0553435     -0.01611948            0.6533265
2  0.45647943  0.19569629      0.0000000     -0.10510424           -0.1099692
3 -0.51426163  0.20612403     -0.7035624     -0.18843917            0.1083179
4  0.00000000 -0.05869012     -1.0553435     -0.26894919           -0.1650872
5 -0.99963216  0.18787549      0.0000000     -0.30991043           -0.1291340
6 -0.09822975  5.31986564      0.0000000     -0.37347098            5.5781823
  Years_Experience
1      -0.24740636
2       1.17446927
3       0.03696877
4       1.26926098
5      -0.24740636
6      -0.15261465
```

Boxplot of SCALED continuous variables

```
boxplot(scaled_data, main = "Boxplots of Scaled Continuous Variables", las = 2, col = "lightgreen")
```



Important; Scaling does NOT remove outliers. It only puts variables on a common scale. The outlier remains an outlier after scaling.

Identify outliers using the IQR method

```
find_outliers <- function(x) {
    Q1 <- quantile(x, 0.25, na.rm = TRUE)
    Q3 <- quantile(x, 0.75, na.rm = TRUE)
    IQR_value <- Q3 - Q1
    lower <- Q1 - 1.5 * IQR_value
    upper <- Q3 + 1.5 * IQR_value
    which(x < lower | x > upper)
}
```

Test

```
find_outliers(data$Income_KES)
```

```
[1]  6 19
```

```
find_outliers(data$Farm_Size_Acres)
```

```
[1] 38 53
```

Count outliers in every continuous variable

```
outlier_count <- sapply(data[continuous_vars], function(x) length(find_outliers(x)))
outlier_count
```

```
                Age           Income_KES       Household_Size 
                  1                    2                    1 
    Farm_Size_Acres Monthly_Expenses_KES     Years_Experience 
                  2                    3                    1 
```

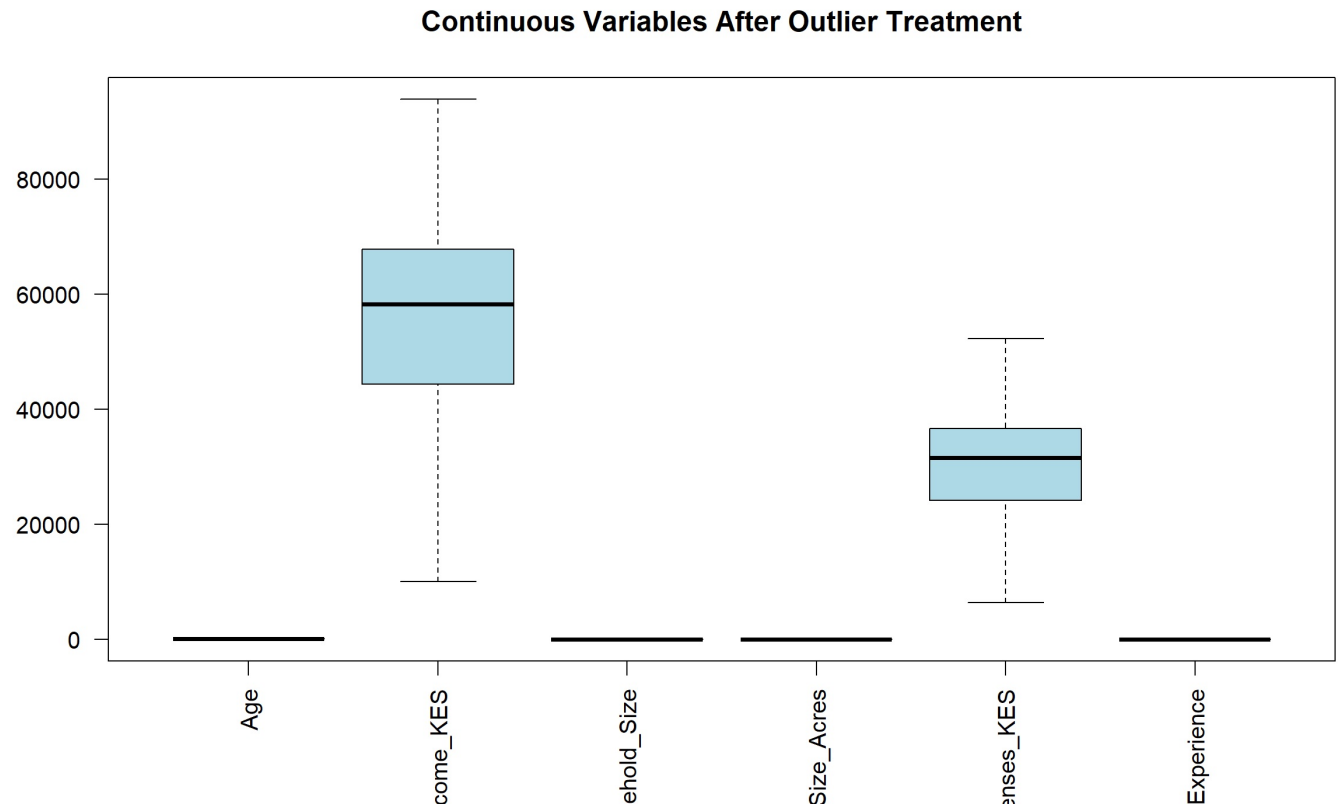
Replace outliers with the MEDIAN

We use the median because it is resistant to extreme observations.

```
replace_outliers_median <- function(x) {
    Q1 <- quantile(x, 0.25, na.rm = TRUE)
    Q3 <- quantile(x, 0.75, na.rm = TRUE)
    IQR_value <- Q3 - Q1
    lower <- Q1 - 1.5 * IQR_value
    upper <- Q3 + 1.5 * IQR_value
    median_value <- median(x, na.rm = TRUE)
    x[x < lower | x > upper] <- median_value
    return(x)
}
for (var in continuous_vars) {
    data[[var]] <- replace_outliers_median(data[[var]])
}
```

Check the boxplots again

```
boxplot(data[continuous_vars], main = "Continuous Variables After Outlier Treatment",
    las = 2, col = "lightblue")
```



Scaled data

```
scaled_data <- as.data.frame(scale(data[continuous_vars]))
boxplot(scaled_data, main = "Boxplots of Scaled Continuous Variables After Outlier Treatment",
    las = 2)
```

Boxplots of Scaled Continuous Variables After Outlier Treatment

